

BHARGAV PATEL

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OBJECTIVE

Computer Science graduate seeking an **Entry-Level Software Engineer** or **Data Analyst/Engineer** role where I can apply my skills in **C++, Python, SQL, and data-driven development** to build reliable systems, analyze insights, and solve real-world problems. Eager to contribute to collaborative engineering teams focused on clean code, scalability, and performance.

PROFESSIONAL SUMMARY

Computer Science graduate experienced in **software development, algorithms, and databases**. Skilled in **C++, Python, Java, SQL, and modern web technologies**. Known for building scalable applications, debugging complex systems, and collaborating effectively in **Agile teams**. Strong foundation in **system-level programming, data analysis, and automation**. Proven ability to **lead teams**, mentor peers, and apply structured problem-solving to deliver clean, efficient, and maintainable code.

EDUCATION

New Jersey Institute of Technology, Newark, NJ

B.S. in Computer Science | GPA: 3.17/4.0 | Jan 2023 - May 2025

Hudson County Community College, Jersey City, NJ

A.S. in Computer Science | GPA: 3.68/4.0 | Jan 2020 - Dec 2022

CERTIFICATIONS & ACHIEVEMENTS

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- Certificate of Accomplishment – Software Engineer (HackerRank)
 - 3rd Place – NJIT Senior Capstone Project Showcase (VR Pedestrian Simulation)
 - NSLS Leadership Certification
 - Dean's List – Hudson County Community College

TECHNICAL SKILLS

Languages: C++, Python, Java, C, SQL, JavaScript, HTML/CSS, Bash

Data Tools & Libraries: Pandas, NumPy, Scikit-learn, Matplotlib

Frameworks & Tools: Git, GitHub, Unity (C#), Jupyter, VSCode, MongoDB, Arduino IDE, TinkerCAD

Databases: MySQL, MongoDB, PL/SQL

Concepts: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Agile/Scrum, Debugging, Linux CLI

Other: Version Control, API Integration, Data Visualization.

PROJECTS

AI Chat Bot Web App – Local Community Chat | HTML, CSS, JavaScript

- Built a simple chat interface where local users can interact with an AI bot on a shared web page.
- Designed chat UI and message history using dynamic DOM rendering in JavaScript.
- Demonstrated ability to connect UI with backend logic (I'll refine based on your code once you upload it).

Pricing & Features Website (Multi-Page, SCSS-Driven Design) | HTML, SCSS

- Developed a 5-page responsive website with sections for features, pricing, contact, and premium.
- Enhanced UI/UX with reusable SCSS mixing, grid/flex box layouts, and styled tooltips.
- Designed visually consistent page bottoms to avoid blank sections and improve user engagement.
- Practiced responsive design principles to ensure accessibility across desktop and mobile.

VR Pedestrian Safety Simulation (Capstone – 3rd Place Showcase) | Unity, C#

- Led a 4-member team to design a VR simulation for pedestrian safety with KATVR treadmill hardware.

- Integrated immersive interactions, backend logic, and hardware/software synchronization.
- Earned 3rd place showcase award for technical innovation and presentation impact.
- Learned to manage project deadlines, task delegation, and cross-functional collaboration.

Real-Time Chat Web App | HTML, CSS, JavaScript, Node.js, Socket.io

- Built a browser-based chat app so multiple users can talk in real time.
- Implemented WebSockets using Socket.io for instant two-way communication (no page refresh required).
- Created typing indicators and broadcast messaging so users see when someone else is typing.
- Used Express to serve static files and manage server events.

Compiler Construction Projects | C++

- Built a lexical analyzer, parser, and interpreter for a Pascal-like language.
- Implemented recursive-descent parsing, semantic checks, and runtime error handling.
- Gained experience in system-level programming, debugging, and compiler design.
- Improved understanding of how programming languages work from syntax to execution.

Cybersecurity Lab Project – Penetration Testing & IDS | Linux, Nmap, Snort, Security Onion

- Simulated attacks (port scanning, DoS, exploitation) using Nmap, Nessus, Metasploit.
- Configured IDS with Snort rules, Security Onion, and SGUIL for threat detection.
- Practiced penetration testing, security reporting, and incident response workflows.
- Gained appreciation for secure coding practices and defensive software engineering.

Machine Learning Projects | Python, Scikit-learn, Pandas, Matplotlib

- Credit Card Categorization: Classified scores using Decision Tree, KNN, Naive Bayes.
- Weather Predictor: Built regression models with feature scaling and interpretability.
- Salary Prediction: Predicted income levels using Random Forest ensembles.
- Applied data preprocessing, model evaluation, and visualization techniques.
- Strengthened data-driven problem-solving and applied ML concepts to real-world datasets.

Algorithms & Optimization | Python

- Solved the 15-puzzle problem with an efficient A* search algorithm.
- Implemented Huffman coding for encoding, decoding, and compression.
- Practiced dynamic programming and algorithmic problem-solving in academic projects.
- Reinforced critical thinking and efficiency analysis in algorithm design.

I2C Bit-Banged Communication Using PCF8574 | Arduino, Embedded C, I2C

- Implemented manual (bit-banged) I2C communication on Arduino Uno without Wire.h, interfacing PCF8574 I/O expander
- Designed I2C start/stop conditions, bit-level read/write operations, and ACK/NACK handling via direct SDA/SCL control
- Controlled 4 LEDs and read 4 push buttons via PCF8574 quasi-bidirectional pins with reliable input-to-output mapping
- Debugged timing, pin direction, and pull-up resistor issues and validated stable behavior using TinkerCAD simulation

3D Unity Game Development | Unity, C#

- Built multiple interactive 3D games featuring physics, animations, and custom logic.
- Designed playable levels with attention to UI/UX and immersive user experience.
- Earned Unity Certification, validating proficiency in 3D and VR development.
- Applied creative problem-solving to deliver engaging gameplay and smooth mechanics.

EXPERIENCE

Hudson County Community College *Enrollment Assistant*

Jersey City, NJ, USA *Mar 2022 - June 2022*

- Assisted 100+ students with admissions and transcript requests.
- Partnered with administrators to streamline student support and database accuracy.
- Demonstrated leadership through task coordination and process improvements.
- Strengthened communication, organization, and record management skills.